

NEWBOLD, PA, USA

WMO: 997698

Lat: 40.137N

Lon: 74.752W

Elev: 59

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.5	(c) 17.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 24.8	(k) 24.8	(l) 22.6	(m) 25.4	(n) 9.3	(o) 310	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.8	(c) 91.3	(d) N/A	(e) 88.7	(f) N/A	(g) 86.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 7.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.3	(b) 16.8	(c) 14.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.0	32.9	35.1	42.8	54.7	64.9	72.7	78.0	75.6	69.1	58.4	46.9	40.0		
	DBStd	17.09	9.59	8.58	8.29	7.70	6.97	5.65	4.53	4.25	5.73	7.34	7.20	8.30		
	HDD50	1729	534	425	252	33	0	0	0	0	14	142	329			
	HDD65	4515	995	838	689	326	89	7	0	1	24	227	544	775		
	CDD50	3933	4	7	28	173	463	681	868	793	574	275	48	19		
	CDD65	1245	0	0	0	16	87	238	403	329	149	23	0	0		
	CDH74	9421	0	0	1	146	614	1757	3608	2359	837	99	0	0		
	CDH80	3026	0	0	0	33	138	546	1381	709	204	15	0	0		
(14) Wind	WSAvg	5.9	7.1	6.5	6.9	6.5	5.9	5.5	5.0	4.6	4.7	5.7	6.1	6.3		
(15) Precipitation	PrecAvg	42.3	3.1	3.0	3.6	3.1	3.4	3.4	4.9	3.9	4.2	2.8	3.1	3.7		
	PrecMax	59.4	9.0	5.2	7.3	6.6	6.2	6.3	13.4	9.6	9.1	5.4	7.8	7.0		
	PrecMin	30.4	0.4	1.0	0.9	0.8	0.2	0.6	1.3	0.5	0.8	0.0	0.3	0.6		
	PrecStd	7.6	1.9	1.2	1.5	1.4	1.7	1.8	3.2	2.4	2.3	1.4	2.0	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	65.7	71.5	84.9	88.2	92.6	96.1	92.3	89.8	81.5	69.7	67.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	58.2	58.6	65.6	78.2	84.0	88.9	92.7	88.9	85.5	76.6	65.8	61.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	52.2	51.6	60.2	73.2	80.4	86.0	89.6	86.5	81.9	72.5	61.7	57.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	47.1	46.7	55.8	68.6	77.4	82.8	87.2	84.2	78.7	69.2	58.5	52.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	13.8	15.3	17.7	16.9	16.1	15.8	14.8	15.7	14.6	14.5	12.2		
		MCDBR	20.7	21.4	22.4	24.2	20.8	18.5	18.4	17.2	19.2	18.4	20.2	17.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.321	0.359	0.389	0.436	0.500	0.497	0.490	0.404	0.361	0.329	0.309		
	taud		2.466	2.431	2.385	2.329	2.226	2.088	2.111	2.118	2.332	2.463	2.510	2.524		
	Ebn at Noon		271	284	283	281	268	251	250	248	265	266	262	263		
	Edh at Noon		25	30	35	39	44	51	49	48	36	29	24	22		
(44) All-Sky Solar Radiation	RadAvg		609	897	1188	1535	1701	1866	1906	1646	1371	973	691	528		
	RadStd		53	96	96	137	195	158	91	108	145	100	61	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+122

Nomenclature: See separate page